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#include <iostream>
#include <sstream>
#include <string>
using namespace std;
int main() {
    string ip;
    cout << "Enter an IP address (format: x.x.x.x): ";
    cin >> ip;
    stringstream ss(ip);
    string octet;
    int firstOctet;
    getline(ss, octet, '.');
    firstOctet = stoi(octet);
    cout << endl;
    cout << "--- Results ---" << endl;
    cout << "IP Address      : " << ip << endl;
    if (firstOctet >= 1 && firstOctet <= 126) {
        cout << "Class          : A" << endl;
        cout << "Subnet Mask     : 255.0.0.0" << endl;
        cout << "No. of Networks : 2^7 (128 possible networks)" << endl;
        cout << "Usable Hosts    : 2^24 - 2 (16,777,214)" << endl;
        cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
        cout << "Last IP        : " << firstOctet << ".255.255.255" << endl;
    }
    else if (firstOctet == 127) {
        cout << "Class          : Reserved / Loopback" << endl;
        cout << "127.x.x.x is used for troubleshooting." << endl;
    }
    else if (firstOctet >= 128 && firstOctet <= 191) {
        cout << "Class          : B" << endl;
        cout << "Subnet Mask     : 255.255.0.0" << endl;
        cout << "No. of Networks : 2^14 (16,384 possible networks)" << endl;
        cout << "Usable Hosts    : 2^16 - 2 (65,534)" << endl;
        cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
        cout << "Last IP        : " << firstOctet << ".255.255.255" << endl;
    }
    else if (firstOctet >= 192 && firstOctet <= 223) {
        cout << "Class          : C" << endl;
        cout << "Subnet Mask     : 255.255.255.0" << endl;
        cout << "No. of Networks : 2^21 (2,097,152 possible networks)" << endl;
        cout << "Usable Hosts    : 2^8 - 2 (254)" << endl;
        cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
        cout << "Last IP        : " << firstOctet << ".255.255.255" << endl;
    }
    else if (firstOctet >= 224 && firstOctet <= 239) {
        cout << "Class          : D (Multicast)" << endl;
        cout << "Subnet Mask     : Not applicable" << endl;
        cout << "No. of Networks : Not applicable" << endl;
        cout << "Usable Hosts    : Not applicable" << endl;
        cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
        cout << "Last IP        : " << firstOctet << ".255.255.255" << endl;
    }
    else if (firstOctet >= 240 && firstOctet <= 255) {
        cout << "Class          : E (Experimental)" << endl;
        cout << "Subnet Mask     : Not applicable" << endl;
        cout << "No. of Networks : Not applicable" << endl;
        cout << "Usable Hosts    : Not applicable" << endl;
    }
}

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    cout << "Starting IP      : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP        : " << firstOctet << ".255.255.255" << endl;
else {cout << "Invalid IP address." << endl;}
return 0;}

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PART B

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#include <iostream>
#include <cmath>
#include <string>

```

```

using namespace std;

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```

int main() {
    int o1, o2, o3, o4;
    char dot;
    int cidr;

```

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// 1. Get user input for Base IP and CIDR prefix
cout << "Enter base IP address (e.g., 192.168.10.0): ";
cin >> o1 >> dot >> o2 >> dot >> o3 >> dot >> o4;

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cout << "Enter CIDR prefix (e.g., 26 for /26): ";
cin >> cidr;

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// Validate for Class C subnetting range
if (cidr < 24 || cidr > 30) {
    cout << "Error: This program currently supports Class C subnetting (CIDR 24 to 30)."
<< endl;
    return 1;
}

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// 2. Calculate the core subnetting details
int hostBits = 32 - cidr;
int blockSize = pow(2, hostBits);
int numSubnets = 256 / blockSize;
int usableHosts = blockSize - 2;
int lastOctetMask = 256 - blockSize;

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// Display general subnet information
cout << "\n--- General Subnet Information ---" << endl;
cout << "Calculated Subnet Mask : 255.255.255." << lastOctetMask << endl;
cout << "Total Usable Hosts      : " << usableHosts << " per subnet" << endl;
cout << "Total Subnets Created : " << numSubnets << "\n" << endl;

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// 3 & 4. Calculate and Display specifics for each subnet
for (int i = 0; i < numSubnets; ++i) {

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int network = i * blockSize;
int firstHost = network + 1;
int lastHost = network + usableHosts;
int broadcast = network + blockSize - 1;

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cout << "=====" << endl;
cout << "Subnet " << (i + 1) << endl;
cout << "=====" << endl;
cout << "Network Address  : " << o1 << "." << o2 << "." << o3 << "." << network <<
endl;
cout << "First Host      : " << o1 << "." << o2 << "." << o3 << "." << firstHost << endl;
cout << "Last Host       : " << o1 << "." << o2 << "." << o3 << "." << lastHost << endl;
cout << "Broadcast Address : " << o1 << "." << o2 << "." << o3 << "." << broadcast <<
endl;

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// Print all usable IP addresses in this subnet
cout << "\nUsable IP Addresses:" << endl;
for (int j = firstHost; j <= lastHost; ++j) {
    cout << " " << o1 << "." << o2 << "." << o3 << "." << j << endl;
}
cout << endl;
}

```

```

return 0;
}

```

GITHUB link

<https://github.com/sakshamsinghsisodiya1204/computer-network-s-laboratory->